Government of Tamilnadu

Department of Employment and Training

Course : TNPSC Group I Mains Material

Subject : Role and impact of Science and Technology in the development of India

Topic : Advancements in the fields of I.T.

© Copyright

The Department of Employment and Training has prepared the TNPSC Group-I Preliminary and Main Exam study material in the form of e-content for the benefit of Competitive Exam aspirants and it is being uploaded in this Virtual Learning Portal. This e-content study material is the sole property of the Department of Employment and Training. No one (either an individual or an institution) is allowed to make copy or reproduce the matter in any form. The trespassers will be prosecuted under the Indian Copyright Act.

It is a cost-free service provided to the job seekers who are preparing for the Competitive Exams.

Commissioner,

Department of Employment and Training.

ADVANCEMENT IN THE FIELD OF INFORMATION TECHNOLOGY

Introduction

Information technology (IT) is the use of any computers, storage, networking and other physical devices, infrastructure and processes to create, process, store, secure and exchange all forms of electronic data.

The Indian Information Technology (IT) – Information Technology Enabled Services (ITES) sector has registered tremendous growth over the past decade, achieving iconic status all over the world and a reputation for reliable and cost-effective delivery of IT services. India is the leading sourcing destination across the world, accounting for approximately 55 per cent market share of the US$ 185-190 billion global services sourcing business in 2017-18. India's IT & ITES Industry grew to US$ 167 billion in 2017-18. Exports from the industry increased to US$ 126 billion in FY18 while domestic revenues (including hardware) advanced to US$ 41 billion. The industry employs about 10 million workforces, The industry has led the economic transformation of the country and altered the perception of India in the global economy. The IT industry has also created significant demand in the Indian education sector, especially for engineering and computer science. The Indian IT and ITES Industry is divided into four major segments – IT services, Business Process Management (BPM) software products and engineering services, and hardware.

IT sector now has diversified into more lucrative digital segments such as cloud computing, Artificial intelligence (AI) and Big Data. India's Technology and BPM (Business Process Management sector including hardware) is likely to generate revenues of US$ 167 billion during FY 18 comparted to US$ 129 billion in FY16. It is expected to reach $350 billion by 2025. The contribution of the IT sector to India's GDP rose to approximately 8 per cent in FY 18 from 1.2 per cent in FY98.

DEPARTMENT OF ELECTRONICS & INFORMATION TECHNOLOGY (DEITY)

Department of Electronics and Information Technology (Deity) in the Ministry of Electronics and Information Technology is responsible for the formulation, implementation and review of national polices in the field of Information Technology, Electronics and the Internet.

The Mission is to promote e-Governance for empowering citizens, promoting the inclusive and sustainable growth of the Electronics and IT & ITeS industries.

The Vision of the Department coincides with the overarching vision outlined under the Digital India-programme of making digital infrastructure as a utility to every citizen, Governance and Services on demand and digital Empowerment of citizens.

The Department aims to promote electronics hardware manufacturing and-TeS industry and provide-infrastructure for delivery of e-services. It promotes the use of ICT for more inclusive growth. It also promotes information Technology education and information technology-based education.

NATIONAL CYBER COORDINATION CENTRE (NCCC)

Ministry of Electronics and Information Technology has recently made the first phase of National Cyber Coordination Centre (NCCC) operational. NCCC has been set up to scan internet traffic coming into the county to detect the real-time cyber threat and alert various organisations as well as internet service providers for timely action.

CYBER AND INFORMATION SECURITY (CIS) DIVISION

Ministry of Home Affairs has recently created a new division. Cyber and Information Security (CIS) to monitor online crimes and threats, including cyber frauds and hacking, suggest ways to minimize and fight them. This division will track and counter online fraud, hacking, identity theft, dark net, trafficking and cyberattacks on critical information infrastructure.

The CIS Division has four Wings, namely Security Clearance, Cyber Crime Prevention, Cyber Security and Information Security Wings each headed by an Under-Secretary

level Officer. Under this, Indian Cyber Crime Coordination Centre (14C) has been set up to act as a nodal point in the fight against cybercrime. The raising of the Cyber Police Force has also been proposed.

NATIONAL CRITICAL INFORMATION INFRASTRUCTURE PROTECTION CENTRE (NCIPC)

National Critical information Infrastructure Protection Centre (NCIPC) is an organisation of the Government of India created under Sec 70A of the Information Technology Act, 2000 (amended 2008), through a gazette notification on 16th Jan 2014 based in New Delhi, India. It is designated as the National Nodal Agency in respect of Critical Information Infrastructure Protection.

NCIPC will function under the National Technical Research Organization; a technical intelligence gathering agency managed directly by the National Security Advisor in Prime Minister's Office (PMO)

THE INFORMATION TECHNOLOGY ACT, 2000

The Information Technology Act, 2000 (also known as the IT Act). It is the primary law in India dealing with cybercrime and electronic commerce. It is based on the United Nations Model Law on Electronic Commerce 1996 (UNCITRAN Model) recommended by the General Assembly of the United Nations by a resolution dated 30 January 1997.

OBJECTIVES OF THE IT ACT

1. To provide legal recognition for transactions:
2. Carried out using electronic data interchange, and other means of electronic communication commonly referred to as "electronic commerce".
3. To facilitate electronic filing of documents with Government agencies and E-Payments.
4. To amend the Indian Penal Code, Indian Evidence Act, 1872, the Banker's Books Evidence Act 1891, Reserve Bank of India Act, 1934.

IT ACT 2000 AND DATA PROTECTION:

Under Section **43A** of the (Indian) Information Technology Act, 2000, a body corporate who is possessing, dealing or handling any sensitive personal data or information, and is negligent in implementing and maintaining reasonable security practices resulting in wrongful loss or wrongful gain to any person, then such body corporate may be held liable to pay damages to the person so affected. It is important to note that there is no upper limit specified for the compensation that can be claimed by the affected party in such circumstances.

Under **Section 72A** of the (Indian) Information Technology Act, 2000, the disclosure of information, knowingly and intentionally, without the consent of the person concerned and in breach of the lawful contract has also been made punishable with imprisonment for a term extending to three years and fine extending to INR 5,00,000.

As of now, the issue of data protection is generally governed by the contractual relationship between the parties, and the parties are free to enter into contracts to determine their relationship defining the terms personal data, sensitive personal data, data which may not be transferred out of or to India and mode of handling of the same.

It is to be noted that **Section 69** of the Act, which is an exception to the general rule of maintenance of privacy and secrecy of the information, provides that where the Government is satisfied that it is necessary for the interest of:

1. the sovereignty or integrity of India,
2. defence of India.
3. security of the State,
4. friendly relations with foreign States or
5. pubic order or
6. for preventing incitement to the commission of any cognizable offence relating to above or
7. for investigation of any offence,

I may by order, direct any agency of the appropriate Government to intercept, monitor or decrypt or cause to be intercepted or monitored or decrypted any information generated, transmitted, received or stored in any computer resource. This section

empowers, the Government to intercept, monitor or decrypt any information including information of personal nature in any computer resources.

Where the information is such that it ought to be divulged in public interest, the Government may I require disclosure of such information. Information relating to an-national activities which are against national security, breaches of the law or statutory duty or fraud may come under this category.

NATIONAL POLICY ON INFORMATION TECHNOLOGY 2012

The Policy aims to leverage Information & Communication Technology (ICT) to address the country's economic and developmental challenges. The policy is rooted in the conviction that ICT has the power to transforms the lives of people. ICT and Electronics are contributing significantly to the Indian economy, " society and governance. IT is a key driver of the knowledge-based global economy. The right polices and investment in infrastructure can strengthen and enhance India's position as a global IT power house. Use of IT can transforms our economy, enhance equity and lead to improvement in development indices. Thrust areas of the policy:

1. To increase revenues of IT and ITES (Information Technology Enabled Services) Industry from 100 Billon USD to 300 Billion USD by 2020 and expand exports from 69 Billion USD to 200 Billion USD by 2020.
2. To gain significant global market-share in emerging technologies and Services.
3. To promote innovation and R&D-in cutting-edge technologies and the development of applications and solutions in areas like localization, location-based services, mobile value-added services, Cloud Computing, Social Media and Utility models.
4. To encourage the adoption of ICTs in key economic and strategic sectors to improve their competitiveness and productivity.
5. To provide fiscal benefits to SMEs and Start-ups for the adoption of IT in value creation
6. To create a pool of 10 million additional skilled manpower in ICT.
7. To make at least one individual in every household e-literate.

8. To provide for mandatory delivery of and affordable access to all public services in electronic mode.
9. To enhance transparency, accountability, efficiency, reliability and decentralization in Government and in particular, in the delivery of public services.
10. To leverage ICT for key Social Sector initiatives like Education, Health, Rural Development and Financial Services to promote equity and quality.
11. To make India the global hub for development of language technologies, to encourage and facilitate the development of content accessible in all Indian languages and thereby help bridge the digital divide.
12. To enable access to content and ICT applications by differently-abled people to foster inclusive development.
13. To leverage ICT for expanding the workforce and enabling life-long learning.
14. To strengthen the Regulatory and Security Framework for ensuring a Secure and legally complaint Cyberspace ecosystem.
15. To adopt open standards and promote open source and open technologies.

The policy attempts to optimally leverage India's global edge in ICT to advance national competitiveness in other sectors, particularly those of strategic and economic importance. The policy will promote an inclusive and equitable society. The Policy is oriented towards the use of ICT to consciously promote decentralization and empowerment of citizens.

NATIONAL CYBER SECURITY POLICY 2013

National Cyber Security Policy is a policy framework by Department of Electronics and Information Technology (Deity). It aims at protecting the public and private infrastructure from cyber-attacks. The policy also intends to safeguard 'information, such as personal information (of web users), financial and banking information and sovereign data'. Due to the dynamic, nature of cyberspace, there was a need for a National cyber Security Policy, with an Integrated vision and a set of sustained and coordinated strategies for securing the cyberspace. It serves as an umbrella framework for defining and guiding the actions related to the security of cyberspace. It also enables the individual sectors and organizations in designing appropriate cybersecurity

policies to suit their needs. The policy provides an overview of what it takes to effectively protect information, information systems and networks and also gives an insight into the Government's approach and strategy for the protection of cyberspace in the county. This policy, therefore, aims to create a cybersecurity framework, which leads to specific action and programmes to enhance the security posture of the country's cyberspace.

Vision: To build secure and resilient cyberspace for citizens, business, and government and also to protect anyone from intervening in into someone's privacy.

Mission: To protect information and information infrastructure in cyberspace, build capabilities to prevent and respond to the cyber threat, reduce vulnerabilities and minimize damage from cyber incidents through a combination of institutional structures, people, processes, technology, and cooperation.

Objective:

Ministry of Communications and Information Technology (India) define the objectives of the policy are as follows:

1. To create a secure cyber ecosystem in the country, generate adequate trust and confidence in the IT system and transactions in cyberspace and thereby enhance adoption of IT in all sectors of the economy.
2. To create an assurance framework for the design of security policies and promotion and enabling actions for compliance to global security standards and best practices by way of conformity assessment (Product, process, technology & people).
3. To strengthen the Regulatory Framework for ensuring a SECURE CYBERSPACE ECOSYSTEM,
4. To enhance and create National and Sectoral level 24X7 mechanism for obtaining strategic information regarding threats to ICT infrastructure, creating scenarios for response, resolution and actions. Crisis management through effective predictive, preventive, protective response and recovery action.
5. To improve the visibility of integrity of ICT products and services by establishing infrastructure for testing & validation of security of such product.

6. To create a workforce for 500,000 professionals skilled in the next five years through capacity building skill development and training.
7. To provide fiscal benefit to businesses for the adoption of standard security practices and processes.
8. To enable Protection of information while in the process, handling, storage & transit to safeguard the privacy of citizen's data and reducing economic losses due to cybercrime or data theft.
9. To enable effective prevention, investigation and prosecution of cybercrime and enhancement of law enforcement capabilities through appropriate legislative intervention.

Strategies

1. Creating a secured Ecosystem.
2. Creating an assurance framework.
3. Encouraging Open Standards.
4. Strengthening the Regulatory Framework.
5. Creating a mechanism for Security Threats Early Warming, Vulnerability management and response to a security threat.
6. Securing E-Governance services.
7. Protection and resilience of Critical Information Infrastructure.
8. Promotion of Research and Development in cyber security.
9. Reducing supply chain risks
10. Human Resource Development (fostering education and training programs both in formal and informal sectors to support Nation's cyber security needs and build capacity.
11. Creating cyber security awareness.
12. Developing an effective Public Private Partnership.
13. To develop a bilateral and multilateral relationship in the area of cyber security with another country. (Information sharing and cooperation)
14. Prioritized approach for implementation.

ISSUES AND UPDATION:

India's cyber landscape has witnessed growing digitization as part of the Governments Digital India push, as well as more sophisticated cyber threats, particularly the WannaCrypt and Petya ransomware attacks that hit Indian networks previous year. These radical changes necessitate a revision and update to India's policy on Cyber Security.

The government of India needs an updated policy to move beyond simply a statement of principles and outline how to operationalize cyber security, from training cybersecurity personnel, to establishing public-private partnerships, and to facilitating civil-military collaboration.

The National Cyber Security Policy broadly outlined a vision for “To create a workforce of 500,000 Professionals skilled in cyber security in the next five years through capacity building, skill development and training" in 2013. After nearly several years, the number for such skilled personnel is only 50,000 or 10% according to latest reports. An updated Cyber Security Policy should outline specific guidelines for the trainings and recruitment of such cyber specialists in a time-bound manner.

Public-private partnerships were a central feature of India's cyber policy as well. The policy called for the "developing effective public-private partnerships and collaborative engagements through technical and operational cooperation and contribution for enhancing the security of cyberspace."

However, there has been little development in this space. Industry partners such as the Information Systems Audit and Control Association (ISACA), the National Association of Software and Services Companies (NASSCOM), and the Data Security Council of India (DSCI) have collaborated to address private sector cyber security needs, but these processes have not yet aligned with government efforts. Addressing this gap must be at the heart of the government's updated policy.

Questions:

1. Explain about National Cyber Security Policy – 2013.
2. Discuss how the field of IT advanced nowadays in India.
3. Explain about National Policy and Information Technology.